MoleculeLens: Interpretable Drug–Text Retrieval via Salient Molecular Substructures

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Drug–text retrieval systems can connect molecular structures to mechanism descriptions, but most current models are difficult to interpret mechanistically. They often rely on large black-box encoders, and benchmark scores can be inflated by memorized scaffolds, drug names, or target metadata. As a result, it is unclear whether a retrieved match is supported by meaningful molecular substructures or by easier lexical shortcuts. We present *MoleculeLens*, an interpretable drug–text retrieval framework that makes each alignment decision traceable to salient molecular features. MoleculeLens aligns fixed molecular fingerprints and frozen biomedical text embeddings using learned linear projection heads. This deliberately constrained architecture is the core methodological contribution: the molecule-side linear map ranks which ECFP4 substructure bits support a retrieval without rerunning the model for every masked bit. On a scaffold-disjoint ChEMBL benchmark, MoleculeLens improves Recall@1 by 9.6 points and MRR by 4.8 points over MolPrompt, while also exposing when performance depends on drug-name or target/action metadata. Beyond retrieval accuracy, MoleculeLens provides pharmacophore-level explanations for correct matches, faithful diagnostics for failed retrievals, and a same-model analysis of how lexical cues change the molecular features that drive a score. These results show that lightweight linear bridges can make drug–text alignment more interpretable and chemically grounded without giving up competitive top-rank retrieval performance.

1 Introduction

Drug–text alignment asks whether molecular structures and mechanism descriptions can be embedded in a common space for retrieval [Edwards et al., 2021, Liu et al., 2023a, Zeng et al., 2022]. The problem is useful for mechanism search, dataset curation, and model validation, but it is hard to trust a high retrieval score by itself: shortcut learning and annotation artifacts can inflate apparent performance [Gururangan et al., 2018, Geirhos et al., 2020]. Drug names, target labels, and scaffold-correlated train/test examples can make a model look accurate even when the molecular evidence behind a match is weak, motivating scaffold-aware evaluation in molecular learning [Bemis and Murcko, 1996, Wu et al., 2018]. Standard contrastive dual encoders can retrieve well, but their scores give little direct evidence about which substructures support a retrieved mechanism [Radford et al., 2021, Edwards et al., 2021].

These risks are especially acute in molecule–language retrieval because the text side often contains exact or near-exact identifiers, including drug names, target names, action labels, and curated mechanism templates from resources such as ChEMBL [Gaulton et al., 2012, Zdrazil et al., 2024]. Prior molecule–language systems evaluate retrieval, generation, or representation quality [Edwards et al., 2021, 2022, Pei et al., 2023, Liu et al., 2023b, Zeng et al., 2022], but their standard outputs are ranked lists or generated text rather than an explanation path from a retrieval score to the molecular substructures supporting it. Thus, even when leakage controls are reported, the field still lacks a way

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

to ask whether a high-scoring drug–mechanism match is driven by pharmacophoric evidence or by lexical contamination.

The strength of MoleculeLens is a deliberately constrained, computationally efficient design that makes drug–text retrieval interpretable at the ECFP4-substructure level. It freezes the molecule and text encoders, learns only thin linear projection heads, and uses the molecule-side linear map as a lens on ECFP4 substructure bits. The goal is not to build the largest drug–language model. The goal is to make each prediction traceable to chemically meaningful fingerprint environments, so shortcut behavior can be inspected at the substructure level.

Contributions.

- **A deliberately interpretable retrieval model.** MoleculeLens keeps the molecular fingerprints and biomedical text encoder fixed and trains only two linear bridges. This design makes each drug–text score directly explainable as ranked ECFP4 fingerprint environments, so the model can show which chemical substructures support a proposed mechanism match.
- **Strong top-rank retrieval under scaffold shift.** We construct a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs and test on 435 molecules whose Bemis–Murcko scaffolds are unseen during training. On this hard split, MoleculeLens achieves the strongest Recall@1 and MRR among the evaluated cross-modal baselines (R@1 0.225, MRR 0.304), improving Recall@1 by 9.6 points over MolPrompt and 17.0 points over KV-PLM.
- **Experimentally validated substructure explanations.** MoleculeLens can identify which fingerprint bits drive a retrieval without rerunning the model for every masked bit. This fast ranking closely matches exact normalized-score gradients (mean Pearson 0.905) and exact leave-one-bit-out score drops (0.947), including failed rank-1 retrievals, showing that the explanations are tied to the learned scoring function rather than illustrative post-hoc overlays.
- **Complementary molecule-side and text-side diagnostics.** On the molecule side, MoleculeLens explains correct matches with pharmacophore-level evidence, separates chemically plausible near misses from opaque errors, and exposes lexical leakage: removing only the drug-name span drops Recall@1 by 70.4% and changes the salient substructures that drive retrieval. On the text side, a contrastive logit lens asks a different question: where in the frozen biomedical text encoder drug-retrievable signal becomes available to the learned bridge.

Relation to prior work. Drug–text and molecule–language models include Text2Mol for retrieval [Edwards et al., 2021], MoleculeSTM for structure–text retrieval and editing [Liu et al., 2023a], MolT5 and BioT5 for molecular language modeling [Edwards et al., 2022, Pei et al., 2023], MolCA for graph–language alignment [Liu et al., 2023b], and KV-PLM for biomedical multimodal pretraining [Zeng et al., 2022]. Graphormer [Ying et al., 2021], Uni-Mol [Zhou et al., 2023], ChemBERTa [Chithrananda et al., 2020], and graph pretraining [Hu et al., 2020] are important molecular representation lines, though not direct text-query retrieval systems without additional alignment. MoleculeLens is complementary to higher-capacity models and molecular contrastive systems such as MolCLR, SMICLR, and CLOOME [Wang et al., 2022, Pinheiro et al., 2022, Sanchez-Fernandez et al., 2023]: it emphasizes scaffold-disjoint drug–mechanism retrieval and substructure-level mechanistic explanations. The evaluation design follows the molecular-learning practice of scaffold splits [Bemis and Murcko, 1996] and combines it with text leakage controls.

MoleculeLens also builds on the dual-encoder retrieval tradition. Scientific and biomedical text encoders such as SciBERT, PubMedBERT, and S-Biomed-RoBERTa provide frozen language representations for domain text [Beltagy et al., 2019, Gu et al., 2021, Deka et al., 2021], while CLIP-style contrastive alignment and InfoNCE-style objectives motivate the shared retrieval-space training recipe [Radford et al., 2021, van den Oord et al., 2018]. Related leave-one-out and Hopfield variants show alternative contrastive objectives for small-data alignment [Poole et al., 2019, Fürst et al., 2022]; MoleculeLens instead uses the linearity of the trained bridge as the route to substructure-level explanation.

The interpretability component is related to sparse and linear feature analysis [Bricken et al., 2023, Templeton et al., 2024] and attribution-graph work [Lindsey et al., 2025], but our features are chemically grounded from the start because ECFP4 bits correspond to circular molecular environments [Rogers and Hahn, 2010] recoverable with RDKit [RDKit contributors, 2023].

2 MoleculeLens Framework

Architecture. Figure 1 summarizes the MoleculeLens architecture. MoleculeLens does not attempt to interpret every hidden unit of a large multimodal model. Instead, it inserts a small linear bridge between a fixed molecular representation and a frozen biomedical text encoder. That bridge creates a shared retrieval space and, because the molecule-side map is linear, a direct route from each retrieval score back to ECFP4 substructures.

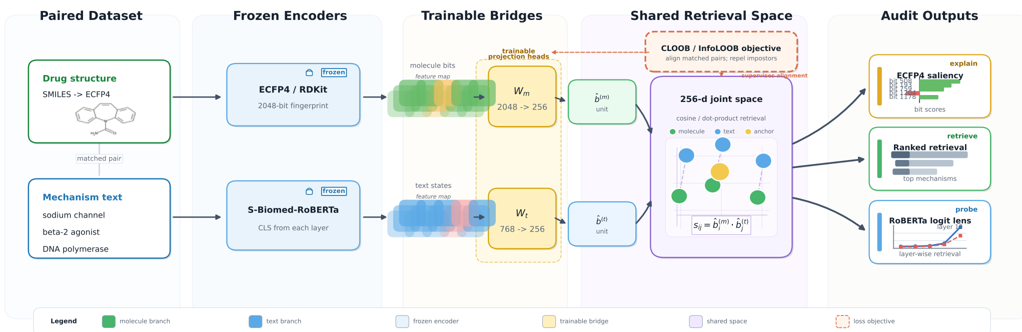

Figure 1: **MoleculeLens architecture for interpretable drug–text retrieval.** A fixed RDKit ECFP4 fingerprint and a frozen biomedical text encoder are connected by two learned linear projection heads trained with a symmetric InfoNCE retrieval objective. Because the molecule-side head is linear, each drug–text score can be traced to ranked ECFP4 fingerprint environments without rerunning the model for every masked bit. The same trained text head is reused as a layer-wise logit lens to identify where drug-retrievable information appears in the frozen text encoder.

Linear bridge retrieval model. For drug i, RDKit computes x_i, a radius-2, 2048-bit ECFP4 fingerprint whose bits encode local molecular environments [Rogers and Hahn, 2010]. For text i, a frozen S-Biomed-RoBERTa sentence encoder [Deka et al., 2021] produces a CLS-style embedding z_i. MoleculeLens learns only two linear projection heads,

$$b_i^{(m)} = W_m x_i + b_m, \qquad b_i^{(t)} = W_t z_i + b_t, \tag{1}$$

followed by L_2 normalization and cosine scoring. Training uses symmetric InfoNCE [van den Oord et al., 2018] with same-target weighting and a hardest-same-target margin term. The scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 epochs. All encoders are frozen; only W_m, b_m, W_t, b_t are trained. Training the projection heads takes under 30 seconds per run on a single NVIDIA GPU; ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes. The dominant cost is the one-time frozen text encoding shared across experiments. Additional implementation details are in Appendix B.

Substructure saliency from the molecular bridge. The normalized retrieval score is

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \mathrm{normalize}(W_m x_i + b_m) \cdot \mathrm{normalize}(W_t z_j + b_t). \tag{2}$$

Because W_m is linear, we use the closed-form ECFP4 saliency proxy

$$\mathrm{attr}_k(i, j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{3}$$

where $\hat{b}_j^{(t)} = \mathrm{normalize}(W_t z_j + b_t)$. Equation 3 is a fast ranking proxy rather than an exact causal decomposition: the exact normalized score also depends on the molecule-side bias and $\|W_m x_i + b_m\|$. This is why the experimental section validates it against exact gradients and exact leave-one-bit-out score drops before using it for chemical interpretation.

Diagnostics enabled by the same bridge. The same linear bottleneck supports two complementary kinds of diagnostic analysis. The first is molecule-side interpretability: saliency over present ECFP4 bits lets us inspect correct and incorrect retrievals as substructure-level decisions, and removing only the explicit drug-name span at inference tests whether lexical leakage changes which molecular features are salient. The second is text-side localization: applying the trained text bridge to each hidden layer of the frozen text encoder does not explain atoms or fingerprint bits, but instead asks where drug-retrievable text information becomes available to the shared retrieval space.

3 Experimental Validation

Dataset and setup. We construct the benchmark from ChEMBL [Gaulton et al., 2012, Zdrazil et al., 2024] by filtering for approved drugs (`max_phase=4`), valid mechanism text, and parseable canonical SMILES. The final dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds. The primary text field concatenates mechanism of action, target name, action type, and drug name:

$$\texttt{text\_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{4}$$

The main split uses seed 0 over unique scaffolds: 2,019 train examples, 245 validation examples, and 435 held-out test examples whose scaffolds are unseen during training. We report text-query retrieval against a molecule gallery using Recall@1, MRR, Recall@5, and Recall@10 for apples-to-apples comparison with baselines. All main-table rows use single-label diagonal evaluation: a prediction is counted correct only when the exact matching row is returned. This is intentionally conservative for drugs with multiple ChEMBL mechanism rows, but it is the only consistent apples-to-apples protocol for baselines where per-example ranked lists are not available.

Several test drugs appear under multiple target–action rows. For attribution analyses, same-drug and wrong-but-close filtering use RDKit `FragmentParent` standardisation over SMILES to avoid treating salt or formulation variants as independent structures. In the canonical scaffold test set, 193 of 435 rows belong to 76 parent structures with multiple mechanism rows.

3.1 Model Performance: Scaffold Retrieval and Leakage Controls

Table 1 shows the main comparison. MoleculeLens has the strongest scaffold-split Recall@1 and MRR among the evaluated cross-modal baselines, improving Recall@1 over MolPrompt by 9.6 points and over KV-PLM by 17.0 points. MolPrompt has higher Recall@5/10, so the claim is not leaderboard dominance: MoleculeLens is strongest at top-rank retrieval while remaining competitive at broader candidate coverage. Full-gallery rows are included only as context; the Thin Bridges global result is a memorization-heavy reference, not the main scientific comparison.

Table 1: **Model performance under scaffold-disjoint retrieval.** All rows use single-label diagonal evaluation, so a query is counted correct only when it retrieves the exact paired molecule row. Full-gallery rows test easier global lookup and are included only as context; scaffold rows test held-out Bemis–Murcko scaffolds and define the main scientific comparison. MoleculeLens achieves the strongest scaffold Recall@1 and MRR, while MolPrompt remains stronger at broader Recall@5/10 coverage.

Model	Eval set	R@1	MRR	R@5	R@10
MoleculeSTM (zero-shot)	Full (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold (N=435)	0.002	0.015	0.011	0.025
MoleculeLens	Scaffold (N=435)	**0.225**	**0.304**	0.384	0.462
MolPrompt	Scaffold (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold (N=435)	0.055	0.121	0.172	0.251

Text-field and leakage controls. Because `text_rich` includes target/action metadata and an explicit drug field, we report three complementary controls in Table 2. Here, "leakage drop" refers to a family of diagnostics rather than a single number: the 51.0% retrained no-drug drop is the fairest model-comparison baseline, the 70.4% same-model no-name drop is the fixed-weight causal diagnostic, and the 22.9% multi-seed proxy mean in Figure 2 measures scaffold-partition sensitivity rather than canonical leakage. Removing only the drug field and retraining gives R@1 = 0.110, a 51.0% relative drop. Holding the rich model fixed and removing only `Drug: X.` at inference gives R@1 = 0.067, a stricter 70.4% same-model causal diagnostic. Training on mechanism text alone, removing target name, action type, and explicit drug name, gives R@1 = 0.124 and MRR = 0.226. Thus target/action fields contribute real signal, but mechanism-only retrieval remains nontrivial and close to MolPrompt scaffold R@1.

Table 2: **Text-field ablations and leakage controls.** Canonical scaffold split. ML denotes MoleculeLens; `text_rich` includes mechanism, target, action, and drug name. The same-model no-name row removes only the explicit drug-name span at inference with weights fixed.

Condition	R@1	MRR	R@1 95% CI
ML `text_rich`	0.225	0.304	[0.186, 0.267]
ML retrained no-drug	0.110	–	[0.083, 0.140]
ML same-model no-name	0.067	–	[0.044, 0.092]
ML mechanism-only	0.124	0.226	–
MolPrompt scaffold	0.129	0.256	[0.101, 0.164]
KV-PLM scaffold	0.055	0.121	[0.037, 0.081]

Table 3: **Full-objective multi-seed scaffold robustness.** Each row retrains MoleculeLens on a different Bemis–Murcko scaffold split using the complete objective. Seeds 0–4 match the proxy robustness analysis; mean and standard deviation show that top-rank performance is not specific to one split.

Seed	n_{test}	R@1	R@5	R@10	MRR
0	403	0.221	0.335	0.424	0.291
1	261	0.295	0.479	0.533	0.381
2	224	0.259	0.375	0.442	0.328
3	221	0.407	0.561	0.629	0.484
4	395	0.203	0.342	0.451	0.284
Mean	–	**0.277**	**0.418**	**0.496**	**0.354**
Std	–	0.081	0.098	0.085	0.083

The non-overlapping Recall@1 intervals for MoleculeLens `text_rich` and MolPrompt support the reliability of the top-rank improvement despite the 435-example test set. Table 3 shows that the complete objective is not weaker than the simplified proxy: mean R@1 is 0.277 ± 0.081 and mean MRR is 0.354 ± 0.083. The higher variance is expected because each seed induces a different scaffold-held-out test set, but every full-loss seed remains above KV-PLM scaffold R@1 and four of five exceed MolPrompt scaffold R@1.

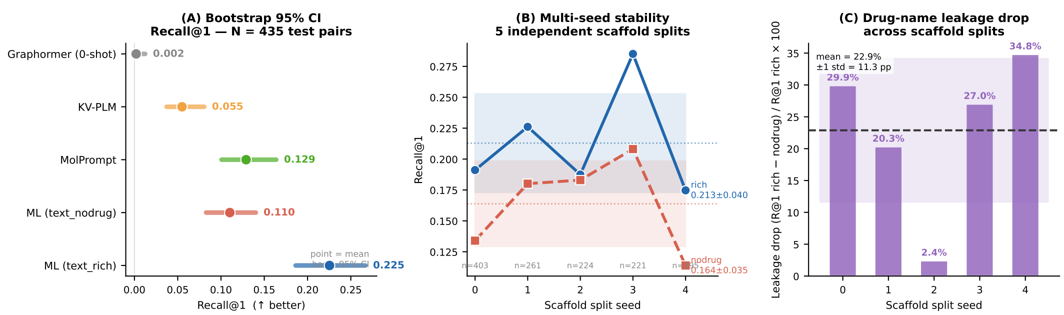

Figure 2: **Robustness of retrieval gains and leakage estimates.** The figure combines bootstrap confidence intervals for canonical scaffold Recall@1, performance variation across five scaffold seeds, and per-seed drug-name leakage drops. Non-overlapping Recall@1 intervals support the reliability of MoleculeLens's top-rank gain over MolPrompt and KV-PLM, while the leakage panel shows that the magnitude of name dependence varies across scaffold partitions.

3.2 Molecule-Side Interpretability: Attribution Faithfulness and Drug-Specific Substructures

We validate Equation 3 directly against exact gradients and exact leave-one-present-bit-out score drops over all 435 scaffold test pairs before using it as an interpretability tool.

Table 4 supports rank-based use of the saliency proxy. It tracks the exact normalized-score gradient strongly and the exact bit-drop score even more closely. The identical-text test addresses a key conceptual concern: the exact top-10 bit-drop sets are almost never identical across distinct drugs sharing the same no-drug query, so the implemented normalized score is strongly drug-specific. The proxy also remains faithful for failed retrievals, which means incorrect predictions remain

Table 4: **Faithfulness of the ECFP4 saliency score.** Validation of Equation 3: all-pair metrics compare fast ECFP4 saliency with exact score-derived references over 435 scaffold test pairs, while stress tests cover failed rank-1 retrievals and distinct drugs sharing identical no-drug text.

All-pair faithfulness	Value	Stress test	Value
Pearson(attr, exact gradient)	0.905	Incorrect rank-1: Pearson	0.970
Pearson(attr, exact bit-drop)	0.947	Incorrect rank-1: top-10 Jaccard	0.761
Top-10 Jaccard(attr, exact bit-drop)	0.710	Same no-drug text: mean Jaccard	0.136
		Same no-drug text: distinct top-10 sets	952/954

interpretable decisions of the learned scoring function rather than cases where the attribution method collapses. Appendix D provides additional case studies.

Across these examples and the additional cases in Appendix D, the highest-saliency fingerprint bits align with known pharmacophoric or mechanism-defining substructures, including steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

What substructures recur? The saliency lens is useful only if it identifies interpretable molecular features rather than a single global shortcut. Appendix Table 7 summarizes the most frequent high-saliency ECFP4 bits across the scaffold test set. Bit 1538 is the most common, appearing in the top-10 for 85 pairs; RDKit decoding maps it to a benzene-ring environment with at least one heteroatom substituent at radius 2, a common pharmacophore anchor across several mechanism families. Frequencies drop sharply after the leading bits, and several frequent bits have near-zero mean signed attribution because they appear in both positive and negative contexts. This pattern supports the intended interpretation: MoleculeLens exposes recurring chemically meaningful substructures, but retrieval decisions are not dominated by one universal fingerprint bit.

Explaining near-miss retrievals. Faithful attribution is useful beyond correctly retrieved examples. Among rank-2/3 errors after parent-structure filtering, 30 of 43 have attribution correlation above 0.5 between the correct drug and the retrieved drug, with mean correlation 0.569. These are not counted as correct in the diagonal retrieval metric, but the analysis distinguishes chemically plausible near misses from arbitrary failures. Figure 4 shows a representative case: sodium phenylacetate is ranked second while sodium phenylbutyrate is ranked first, and the two molecules share the same ammonia-lowering mechanism with highly overlapping salient substructures. This does not make the retrieval correct, but it shows that MoleculeLens can expose when an error is driven by related chemical evidence rather than an opaque embedding artifact.

Same-model leakage attribution. We next ask whether the same rich model uses the same substructures when the explicit drug name is removed at inference. For each test pair, let $\mathcal{T}_i^{\mathrm{rich}}$ and $\mathcal{T}_i^{\mathrm{nodrug}}$ be the top-10 saliency-bit sets under the original query and the same query with `Drug: X.` removed. We compute

$$J_i = \frac{|\mathcal{T}_i^{\mathrm{rich}} \cap \mathcal{T}_i^{\mathrm{nodrug}}|}{|\mathcal{T}_i^{\mathrm{rich}} \cup \mathcal{T}_i^{\mathrm{nodrug}}|}. \tag{5}$$

Mean overlap is only 0.356 ± 0.161; 82/435 pairs have $J_i < 0.2$, 36/435 have $J_i > 0.6$, and 13/435 are leakage candidates that are correct under `text_rich`, wrong under the same-model no-name query, and have $J_i < 0.2$. This converts the score-level leakage drop into a structural claim: for a substantial subset of test pairs, removing the name changes which molecular features drive retrieval even though model weights are fixed.

3.3 Text-Side Diagnostic: Where Retrieval Signal Appears

Section 3.2 asks which molecular substructures support a retrieval, how faithful those saliency rankings are, and how those substructures change under drug-name ablation. This section asks a different question. It does not attribute a retrieval score to chemical substructures. Instead, it probes the frozen text encoder by asking at which layer its CLS representation becomes usable by the already trained MoleculeLens text bridge.

We adapt the logit-lens idea [nostalgebraist, 2020] by applying the trained text projection head W_t to the CLS state at each S-Biomed-RoBERTa layer. The diagnostic exactly replays the main rich model

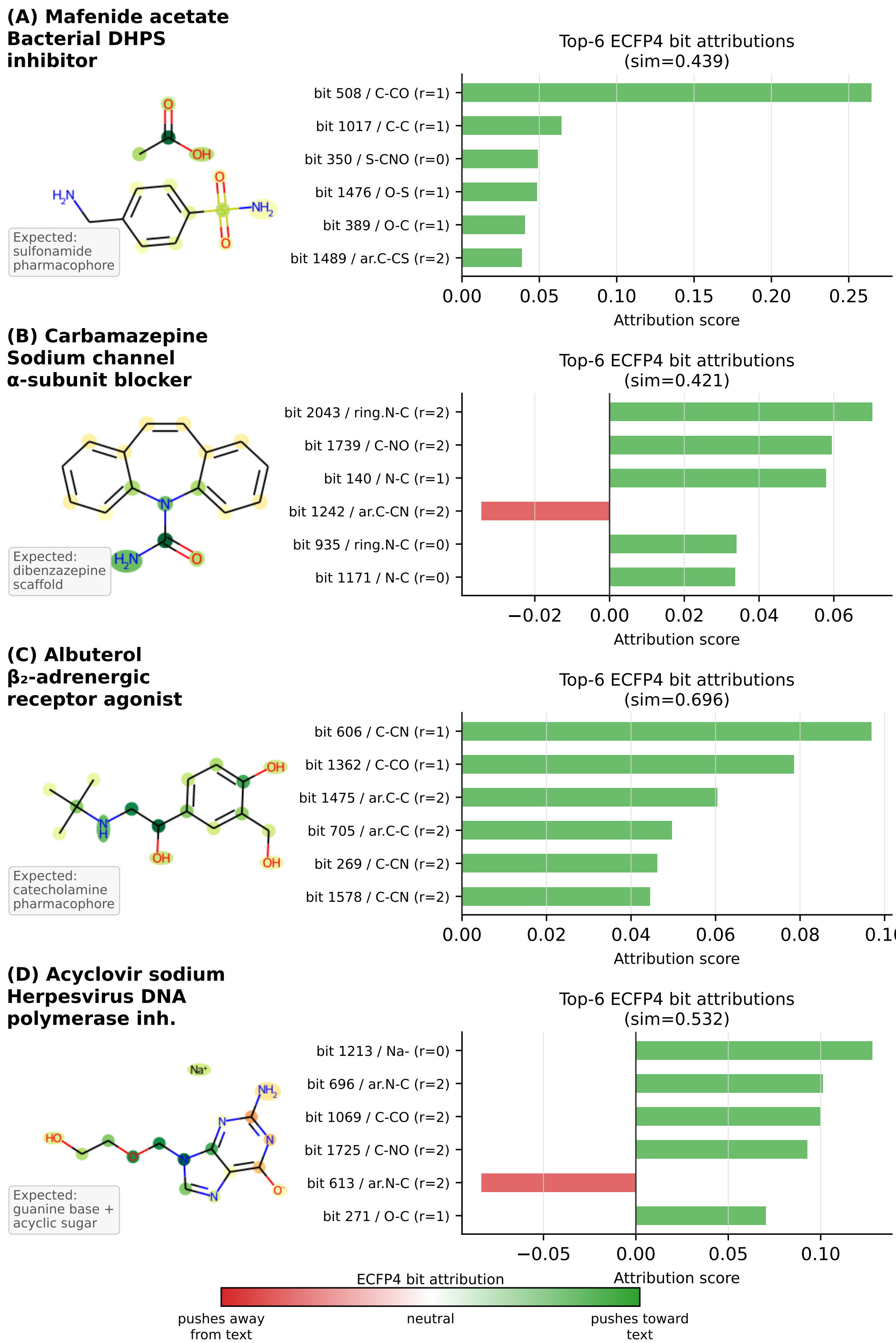

Figure 3: **Molecular-lens explanations for correctly retrieved mechanisms.** Each panel links a successful drug–text retrieval to the ECFP4 fingerprint environments that most support the score. Atoms are colour-coded by signed attribution value and paired with top-bit bar charts, so the reader can see both where the evidence lies on the molecule and which fingerprint bits drive the ranking. Across the shown examples, high-saliency bits align with pharmacophoric or mechanism-defining regions rather than arbitrary fragments.

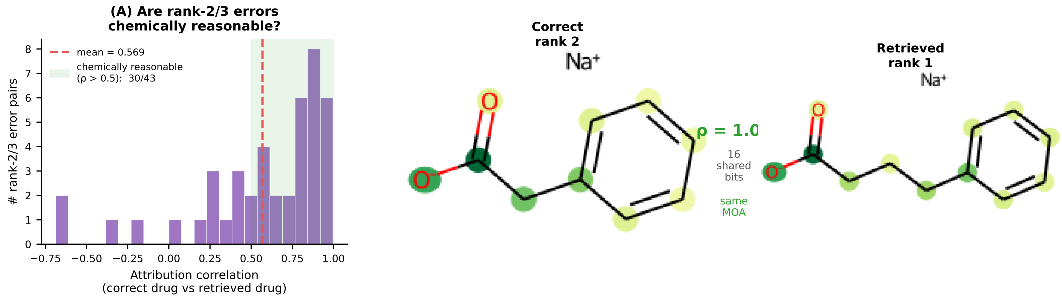

Figure 4: **Near-miss explanations separate chemically plausible errors from opaque failures.** For scaffold-test queries whose correct match is ranked second or third, we compare the saliency vectors of the correct molecule and the top retrieved molecule. High attribution correlation indicates that the model is using overlapping chemical evidence even though the diagonal retrieval metric counts the result as wrong. Sodium phenylacetate retrieved as sodium phenylbutyrate illustrates a high-correlation, same-mechanism confusion.

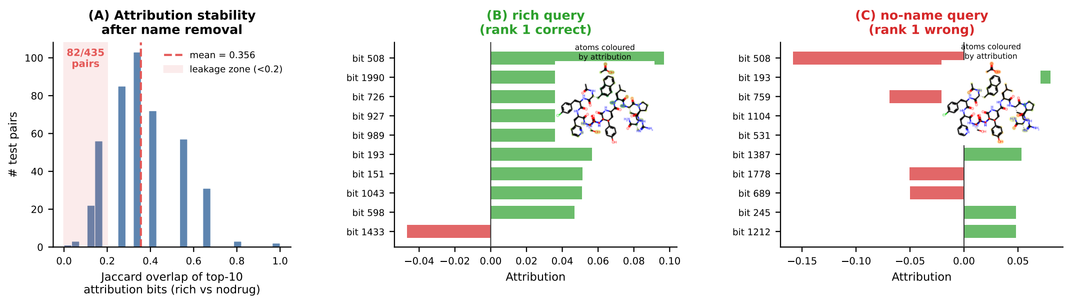

Figure 5: **Drug-name leakage changes the substructures used by the same model.** The rich query includes the explicit drug name, while the no-name query removes only that span at inference with all model weights fixed. Low top-10 ECFP4 saliency overlap means that removing lexical information changes which molecular environments drive retrieval. Cetrorelix acetate illustrates a leakage candidate where the prediction and salient substructures both shift after the name is removed.

at layer 12: the reconstructed embeddings match the saved outputs within 1.21×10^{-3} maximum absolute error and recover R@1 = 0.225. Layers 0–11 are near chance for both rich and retrained no-drug text (R@1 between 0.002 and 0.011), while layer 12 jumps to R@1 = 0.225 for `text_rich` and 0.113 for `text_nodrug`. This is best viewed as a descriptive sanity check: retrieval signal and the rich-vs-no-name gap both appear in the final contextualized CLS representation, not gradually across intermediate layers. The full layer table and attention visualizations are in Appendix E.

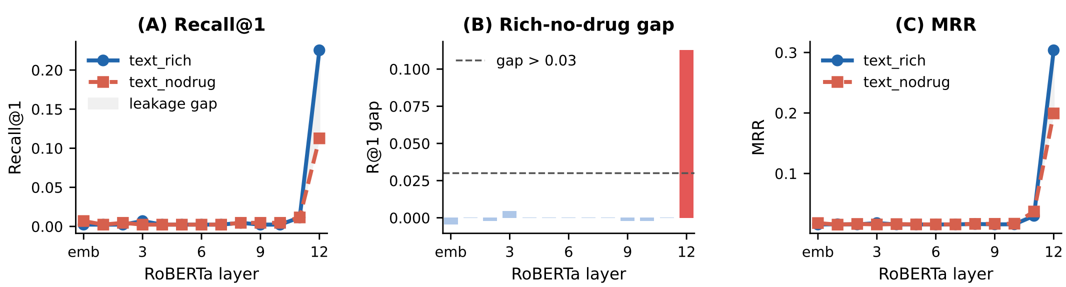

Figure 6: **Text-side logit lens localizes drug-retrievable information to the final RoBERTa layer.** The trained text projection head is applied to CLS embeddings from each frozen S-Biomed-RoBERTa layer and evaluated against the molecule gallery. Retrieval remains near chance through layers 0–11, then layer 12 recovers the main rich-model performance and reveals the gap caused by removing explicit drug names.

Layer-12 retrieval is heterogeneous across mechanism families (Table 5): nuclear-receptor, kinase, antiviral, and antibacterial families show the clearest signal, while GPCR and Other remain harder.

Table 5: **Mechanism families with strongest final-layer text retrieval.** Layer-12 Recall@1 under `text_rich` on the scaffold test set, sorted by family-level R@1.

Mechanism family	n	R@1	Mechanism family	n	R@1
Nuclear receptor agonist/antagonist	18	0.611	Diagnostic agent	15	0.333
Kinase inhibitor	11	0.545	GPCR agonist/antagonist	129	0.271
Antiviral DNA polymerase inhibitor	15	0.533	COX inhibitor	16	0.250
Antibacterial	25	0.440	Ion channel blocker	41	0.195
Other	165	0.152			

This points to curated mechanism text and family-specific lexical structure rather than chemically useful retrieval signal in intermediate RoBERTa layers.

4 Discussion and Limitations

MoleculeLens is not a claim that frozen ECFP4 plus a linear bridge solves scaffold-generalized drug–text retrieval. Its narrower contribution is that a lightweight model remains competitive at top-rank scaffold retrieval while making its decisions inspectable at the ECFP4-substructure level. The saliency proxy is validated against exact score-derived quantities, remains faithful on retrieval failures, and exposes same-model drug-name leakage as a shift in salient molecular features.

The benchmark has important limitations. The rich text field includes target and action metadata, and the mechanism-only ablation confirms that these fields contribute retrieval signal; the rich-text result should therefore be interpreted as curated mechanism-metadata retrieval rather than pure free-text mechanism understanding. Implicit lexical cues, such as proprietary-name fragments embedded in mechanism text, are not removed by the no-drug ablation and may still inflate no-drug scores. The dataset covers FDA-approved drugs with curated ChEMBL mechanisms and does not establish generalization to investigational compounds or non-ChEMBL prose. Finally, the baseline set does not include adapted MolT5, BioT5, Uni-Mol, MolCA, or chemistry-LLM retrieval systems; these require nontrivial task adaptation and remain important future comparisons.

Reproducibility and intended use. The scaffold split, paper-facing tables, and main-number manifests are released to make the benchmark's leakage controls visible rather than implicit. Changing from scaffold-held-out to full-gallery evaluation, or from same-model no-name ablation to retrained no-drug comparison, answers a different scientific question, so we report rich, mechanism-only, and no-name conditions side by side. MoleculeLens is an interpretability tool, not a clinical or regulatory decision system: its use is to show which fingerprint environments support a proposed structure–mechanism match and whether those environments shift under text perturbations.

Future interpretability directions. The shared 256-dimensional embedding space is a natural target for sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] and cross-modal crosscoders [Lindsey et al., 2025] that separate genuine structure–mechanism features from text-only leakage. MoleculeLens leaves these extensions to future work, but supplies the fixed split, saved embeddings, and substructure-level saliency targets needed to test them.

Benchmark scope. The benchmark covers FDA-approved ChEMBL drugs with parseable SMILES and curated mechanism text; it does not cover investigational compounds, unapproved natural products, or out-of-domain mechanism prose. Same-modality text lookup controls access gallery text rather than molecular structure, so we treat them as text-identifiability diagnostics rather than competing molecular-retrieval baselines.

5 Conclusion

On a 2,699-pair ChEMBL benchmark with scaffold-disjoint evaluation, MoleculeLens outperforms MolPrompt and KV-PLM on Recall@1 and MRR while exposing lexical and target/action metadata dependence. Its central value is interpretability: a validated ECFP4 saliency lens explains correct matches, diagnoses failed retrievals, and characterizes same-model leakage, giving future drug–text systems a chemically grounded benchmark for feature-level analysis.

References

Iz Beltagy, Kyle Lo, and Arman Cohan. SciBERT: A pretrained language model for scientific text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 2019.

Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josh E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. `https://transformer-circuits.pub/2023/monosemantic-features/`, 2023.

Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. ChemBERTa: Large-scale self-supervised pretraining for molecular property prediction, 2020.

Pritam Deka, Anna Jurek-Loughrey, and Deepak. Unsupervised keyword combination query generation from online health related content for evidence-based fact checking. In *The 23rd International Conference on Information Integration and Web Intelligence*, pages 267–277, 2021. doi: 10.1145/3487664.3487701.

Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, and Heng Ji. Text2mol: Cross-modal molecule retrieval with natural language queries. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, Kyunghyun Cho, and Heng Ji. Translation between molecules and natural language. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 375–413, 2022. doi: 10.18653/v1/2022.emnlp-main.26.

Andreas Fürst, Elisabeth Rumetshofer, Johannes Lehner, Viet T. Tran, Fei Tang, Hubert Ramsauer, David Kreil, Michael Kopp, Günter Klambauer, Angela Bitto-Nemling, and Sepp Hochreiter. CLOOB: Modern Hopfield networks with InfoLOOB outperform CLIP. In *Advances in Neural Information Processing Systems*, 2022.

Anna Gaulton, Louisa J. Bellis, A. Patricia Bento, Jon Chambers, Mark Davies, Anne Hersey, Yvonne Light, Shaun McGlinchey, David Michalovich, Bissan Al-Lazikani, and John P. Overington. ChEMBL: A large-scale bioactivity database for drug discovery. *Nucleic Acids Research*, 40(D1): D1100–D1107, 2012. doi: 10.1093/nar/gkr777.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020. doi: 10.1038/s42256-020-00257-z.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2021.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112. Association for Computational Linguistics, 2018. doi: 10.18653/v1/N18-2017.

Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay Pande, and Jure Leskovec. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations*, 2020.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L Turner, Craig Citro, Tom Conerly, Adly Templeton, Callum McDougall, et al. On the biology of a large language model. `https://transformer-circuits.pub/2025/attribution-graphs/methods.html`, 2025.

Shengchao Liu, Weili Nie, Chengpeng Wang, Jiarui Lu, Zhuoran Qiao, Ling Liu, Jian Tang, Chaowei Xiao, and Anima Anandkumar. Multi-modal molecule structure–text model for text-based retrieval and editing. *Nature Machine Intelligence*, 5:1447–1457, 2023a.

Zhiyuan Liu, Sihang Li, Yanchen Luo, Hao Fei, Yixin Cao, Kenji Kawaguchi, Xiang Wang, and Tat-Seng Chua. MolCA: Molecular graph-language modeling with cross-modal projector and uni-modal adapter. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 15623–15638, 2023b. doi: 10.18653/v1/2023.emnlp-main.966.

nostalgebraist. Interpreting GPT: the logit lens. `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`, 2020.

Qizhi Pei, Wei Zhang, Jinhua Zhu, Kehan Wu, Kaiyuan Gao, Lijun Wu, Yingce Xia, and Rui Yan. BioT5: Enriching cross-modal integration in biology with chemical knowledge and natural language associations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1102–1123, 2023. doi: 10.18653/v1/2023.emnlp-main.70.

Gabriel A. Pinheiro, Juarez L. F. Da Silva, and Marcos G. Quiles. SMICLR: Contrastive learning on multiple molecular representations for semi-supervised and unsupervised representation learning. *Journal of Chemical Information and Modeling*, 62(17):3948–3960, 2022. doi: 10.1021/acs.jcim.2c00521.

Ben Poole, Sherjil Ozair, Aaron van den Oord, Alexander A. Alemi, and George Tucker. On variational bounds of mutual information. In *Proceedings of the 36th International Conference on Machine Learning*, 2019.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.

RDKit contributors. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2023.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Ana Sanchez-Fernandez, Elisabeth Rumetshofer, Sepp Hochreiter, and Günter Klambauer. CLOOME: Contrastive learning unlocks bioimaging databases for queries with chemical structures. *Nature Communications*, 14(1):7339, 2023. doi: 10.1038/s41467-023-43135-9.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. `https://transformer-circuits.pub/2024/scaling-monosemanticity/`, 2024.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. In *arXiv preprint arXiv:1807.03748*, 2018.

Yuyang Wang, Jianren Wang, Zhonglin Cao, and Amir Barati Farimani. Molecular contrastive learning of representations via graph neural networks. *Nature Machine Intelligence*, 4(3):279–287, 2022. doi: 10.1038/s42256-022-00447-x.

Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S. Pappu, Karl Leswing, and Vijay Pande. Moleculenet: A benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018. doi: 10.1039/C7SC02664A.

Chengxuan Ying et al. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, 2021.

Barbara Zdrazil et al. The ChEMBL database in 2023: A drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.

Zheni Zeng et al. A deep-learning system bridging molecule structure and biomedical text with comprehension comparable to human professionals. *Nature Communications*, 13(1):862, 2022. doi: 10.1038/s41467-022-28494-3.

Gengmo Zhou, Zhifeng Gao, Qiankun Ding, Hang Zheng, Hongteng Xu, Zhewei Wei, Linfeng Zhang, and Guolin Ke. Uni-Mol: A universal 3d molecular representation learning framework. In *International Conference on Learning Representations*, 2023.

A Additional Benchmark and Evaluation Details

The primary retrieval direction is text query to molecule gallery because this is the direction used by the attribution and logit-lens analyses. The diagonal evaluation protocol and parent-structure handling are summarized in the main text.

B Architecture and Implementation Details

The model hyperparameters, runtime, and architecture diagram are summarized in the main text.

C Robustness Details

Table 6: **Proxy-objective scaffold variance and leakage sensitivity across seeds.** Five independent Bemis–Murcko scaffold splits are evaluated with the simplified MoleculeLens proxy objective; only the linear projection heads are retrained for each seed. The table reports retrieval metrics and the relative drug-name leakage drop, $(R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}}$, showing that both performance and leakage depend on the held-out scaffold partition.

Seed	n_{test}	R@1	R@5	R@10	MRR	Leakage drop
0	403	0.191	0.377	0.459	0.287	29.9%
1	261	0.226	0.479	0.544	0.342	20.3%
2	224	0.188	0.393	0.496	0.296	2.4%
3	221	0.285	0.525	0.634	0.402	27.0%
4	395	0.175	0.352	0.446	0.269	34.8%
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9%**
Std	–	0.045	0.070	0.073	0.053	12.6%

The corresponding full-loss five-seed check is reported in the main text in Table 3; the robustness summary figure is shown in Figure 2.

D Additional Attribution Analyses

For each test pair, we compute Equation 3, restrict to ECFP4 bits present in the molecule, rank bits by absolute saliency, and decode top-ranked bits with RDKit. Among eight correctly retrieved examples spanning antibacterial, antiviral, nuclear-receptor, opioid, sulfonamide, and diagnostic-agent mechanisms, the highest-saliency bits correspond to pharmacophoric or mechanism-defining regions such as steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

Figure 3 shows representative correct retrievals. Aggregate high-saliency bit frequencies are reported in Table 7.

The same-model drug-name ablation figure is shown in Figure 5.

E Contrastive Logit-Lens Details

Mechanism-family Recall@1 is reported in the main text in Table 5.

F Broader Impacts

Broader impacts and intended-use considerations are discussed in the main text. In brief, MoleculeLens is an interpretability framework for publicly available ChEMBL drug–mechanism retrieval and is not intended for clinical or regulatory deployment without expert oversight.

NeurIPS Paper Checklist

1. **Claims**

Table 7: **Recurring high-saliency ECFP4 environments across the scaffold test set.** Frequency counts how often each bit appears in the top-10 saliency ranking over 435 test pairs, and mean attribution reports its average signed contribution when present. The leading bits recur across multiple mechanism families, but their signed effects are not uniformly positive, supporting the claim that MoleculeLens exposes recurring chemical motifs rather than one universal shortcut bit.

Bit index	Frequency	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

Table 8: **Layer-wise text retrieval under rich and no-drug queries.** The trained text projection head is applied to CLS states from selected frozen RoBERTa layers and evaluated against the molecule gallery. Layers 0–11 carry essentially no drug-retrievable signal under either text condition; meaningful retrieval appears only at layer 12, where the rich-vs-no-drug Recall@1 gap becomes visible.

Layer	text_rich		text_nodrug		Gap
	R@1	MRR	R@1	MRR	R@1
0	0.002	0.016	0.007	0.018	–0.005
3	0.007	0.018	0.002	0.016	0.005
6	0.002	0.016	0.002	0.016	0.000
9	0.002	0.016	0.005	0.017	–0.002
11	0.011	0.030	0.011	0.037	0.000
12	**0.225**	**0.304**	**0.113**	**0.199**	**0.113**

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state four specific contributions (validated closed-form linear saliency framework, leakage anatomy, descriptive contrastive logit-lens diagnostic, reproducible scaffold benchmark) that are each demonstrated experimentally in the corresponding sections. The scope is explicitly limited to FDA-approved drugs with ChEMBL mechanism text under scaffold-disjoint evaluation.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 4 discusses: (i) dataset scope limited to FDA-approved drugs; (ii) residual implicit lexical leakage not captured by the drug-name ablation; (iii) target/action

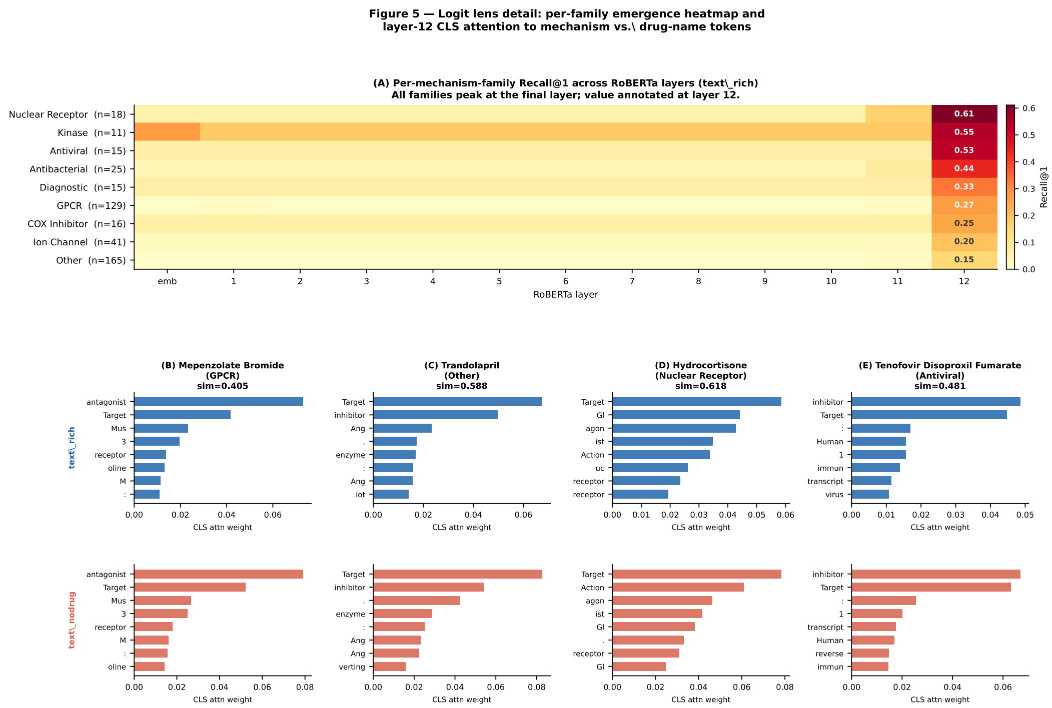

Figure 7: **Detailed text-side diagnostic by mechanism family and attention pattern.** The figure expands the main logit-lens result by showing layer-12 retrieval by mechanism family together with representative CLS attention maps under rich and no-drug text. These panels help distinguish a final-layer retrieval signal from a gradual accumulation of drug-retrievable information across intermediate layers.

metadata leakage quantified by the mechanism-only ablation; (iv) the restriction of the primary comparison to cross-modal structure-aware baselines; (v) missing adapted evaluations of newer molecular language models; (vi) domain-specificity and the open question of generalization to unapproved compounds.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover

limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no formal theorems. Equation 3 is presented as a closed-form linear saliency approximation derived directly from the linear projection heads; its approximation status and omitted bias/normalization terms are stated explicitly in Section 2, so no separate proof is required.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 describes the ChEMBL data construction (max_phase filter, SMILES fetching, Bemis–Murcko scaffold split), Section 2 gives all hyperparameters (batch size, LR, weight decay, temperature, margin, epochs) and states the linear saliency formula and its approximation assumptions, and Appendix B summarizes implementation details. The repo now includes a single refresh command, `bash scripts/run_paper_artifact_refresh.sh`, which rewrites the comparison manifest, canonical Track 1/Track 3 artifacts, paper figures, and a machine-readable `results/paper_artifact_manifest.json` that records the source files and key numbers backing the manuscript.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: Code and benchmark data (scaffold split manifest, projection weights, repo-root README, and the canonical `scripts/run_paper_artifact_refresh.sh` entrypoint) will be released publicly upon acceptance of the paper. The data is derived entirely from ChEMBL, which is publicly available under CC BY-SA 3.0.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3 specifies the Bemis–Murcko scaffold split sizes (2,019 train / 245 val / 435 test), and Section 2 lists all training hyperparameters. The Adam optimizer is used by default; the text encoder is frozen throughout.

   Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 2 reports bootstrap 95% confidence intervals (10,000 resamples) for Recall@1 for MoleculeLens and Wilson score intervals for baselines. Table 3 reports mean and standard deviation of Recall@1 across five independent scaffold splits. The factors of variability (test-set resampling; scaffold partition randomness) are described in Section 3.1 and Figure 2.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 2 states that all experiments run on a single NVIDIA GPU, training completes in under 30 seconds per run, and the attribution and logit-lens analyses each complete in under 10 minutes. The dominant compute cost is the one-time RoBERTa text encoding, which is shared across all experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The research uses publicly available ChEMBL data (CC BY-SA 3.0), does not involve human subjects, does not generate potentially harmful content, and raises no privacy or fairness concerns.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Broader Impacts appendix discusses positive impacts (transparent, interpretable drug-discovery retrieval tools; surfacing evaluation pitfalls) and negative impacts (the framework is not intended for clinical deployment and raises no significant negative risk).

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The model (two small linear projection matrices) and the benchmark (a filtered subset of public ChEMBL data) pose no meaningful misuse risk.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: ChEMBL [Zdrazil et al., 2024] is used under CC BY-SA 3.0. RDKit [RDKit contributors, 2023] is used under the BSD license. S-Biomed-RoBERTa is used for inference only. KV-PLM [Zeng et al., 2022], MolPrompt, and Graphormer [Ying et al., 2021] are cited and evaluated under their respective open-source licenses. All assets are cited in the text.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The new asset is the Bemis–Murcko scaffold-disjoint ChEMBL benchmark (2,699 drug–mechanism pairs with split manifests). Its construction is described in Section 3 and Appendix A. Code and split manifests will be released upon acceptance.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: Does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve human subjects. No IRB approval is required.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: LLMs are not a component of the MoleculeLens method. The text encoder (S-Biomed-RoBERTa) is an encoder-only model used purely for feature extraction without any generative LLM component.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.